LAB ASSIGNMENT-2

DDL QUESTIONS (Data Definition Language)

1. Create table CLASS with appropriate data types

```
INSERT INTO CLASS VALUES
(1, 'Rahul'),
(2, 'Nakul'),
(3, 'Vansh');
```

OUTPUT:

id	name
1	Rahul
2	Nakul
3	Vansh

2. Create table EMPLOYEES with specified attributes

```
CREATE TABLE EMPLOYEES (
    Employee_Id INT PRIMARY KEY,
    First_Name VARCHAR(50),
    Last_Name VARCHAR(50),
    Email VARCHAR(50),
    Phone_Number VARCHAR(20),
    Hire_Date DATE,
    Job_Id VARCHAR(20),
    Salary INT,
    Commission_Pct DECIMAL(5,2),
    Manager_Id INT,
    Department_Id INT
```

OUTPUT:

	Field	Type	Null	Key	Default	Extra
▶	Employee_Id	int	NO	PRI	NULL	
	First_Name	varchar(50)	YES		NULL	
	Last_Name	varchar(50)	YES		NULL	
	Email	varchar(50)	YES		NULL	
	Phone_Number	varchar(20)	YES		NULL	

3.Create table Client_master with specified attributes

```
CREATE TABLE Client_master (
    ClientNo VARCHAR(10) PRIMARY KEY,
    Name VARCHAR(50),
    City VARCHAR(50),
    Bal_due INT
);
```

OUTPUT:

	Field	Type	Null	Key	Default	Extra
▶	ClientNo	varchar(10)	NO	PRI	NULL	
	Name	varchar(50)	YES		NULL	
	City	varchar(50)	YES		NULL	
	Bal_due	int	YES		NULL	

4.Rename table Client_master to Client12

```
RENAME TABLE Client_master TO Client12;
```

OUTPUT:

	Tables_in_college_db
▶	class
	client12
	employees

5.Display bal_due column heading as “BALANCE”

```
SELECT Bal_due AS BALANCE FROM Client12;
```

OUTPUT:

Result Grid

	BALANCE

DML QUESTIONS (Data Manipulation Language)

Part A: Transactions and Savepoints (CLASS Table)

1. Insert + Commit

```
INSERT INTO CLASS VALUES (5, 'Rahul');
COMMIT;
```

OUTPUT:

	id	name
▶	1	Bravo
	2	Alex
	4	Cheng
	5	Rahul
*	NULL	NULL

2. Update + Savepoint A

```
UPDATE CLASS SET name = 'Abhijeet' WHERE id = 5;
SAVEPOINT A;
```

OUTPUT:

id	name
1	Bravo
2	Alex
4	Cheng
5	Abhijeet
NULL	NULL

3. Insert + Savepoint B

```
INSERT INTO CLASS VALUES (6, 'Chris');
SAVEPOINT B;
```

OUTPUT:

id	name
1	Bravo
2	Alex
4	Cheng
5	Abhijeet
6	Chris

4. Insert + Savepoint C

```
INSERT INTO CLASS VALUES (7, 'Bravo');
SAVEPOINT C;
```

OUTPUT:

id	name
4	Cheng
5	Abhijeet
6	Chris
7	Bravo
NULL	NULL

5. Rollback to Savepoint B

```
ROLLBACK TO B;
```

OUTPUT:

id	name
1	Bravo
2	Alex
4	Cheng
5	Abhijeet
6	Chris

6. Rollback to Savepoint A

```
ROLLBACK TO A;
```

OUTPUT:

id	name
1	Bravo
2	Alex
4	Cheng
5	Abhijeet
NULL	NULL

PART B: Retrieving Records (EMPLOYEES Table)

1.Find out the employee id, names, and salaries of all the employees.

```
SELECT employee_id, first_name, last_name, salary
FROM employees;
```

OUTPUT:

employee_id	first_name	last_name	salary
101	Steven	King	24000.00
102	Neena	Kochhar	17000.00
103	Lex	De Haan	17000.00
104	Alexander	Hunold	9000.00
105	Bruce	Ernst	6000.00

2.List out the employees who work under manager 100.

```
SELECT *
FROM employees
WHERE manager_id = 100;
```

OUTPUT:

employee_id	first_name	last_name	email	phone_number	hire_date	job_id	salary	commission_pct	manager_id	department_id
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

3.Find the names of the employees who have a salary greater than or equal to 4800.

```sql
SELECT first_name, last_name
FROM employees
WHERE salary >= 4800;
```

OUTPUT:

first_name	last_name
Steven	King
Neena	Kochhar
Lex	De Haan
Alexander	Hunold
Bruce	Ernst

4.List out the employees whose last name is 'AUSTIN'.

```sql
SELECT *
FROM employees
WHERE last_name = 'Austin';
```

OUTPUT:

employee_id	first_name	last_name	email	phone_number	hire_date	job_id	salary	commission_pct	manager_id	department_id
106	David	Austin	DAUSTIN	5904234569	2005-06-25	IT_PROG	4800.00	NULL	103	60
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

5.Find the names of the employees who work in departments 60, 70, and 80.

```sql
SELECT first_name, last_name
FROM employees
WHERE department_id IN (60,70,80);
```

OUTPUT:

first_name	last_name
Alexander	Hunold
Bruce	Ernst
David	Austin
Valli	Pataballa
Diana	Lorentz

6.Display the unique Manager_Id.

```
SELECT DISTINCT manager_id
FROM employees;
```

OUTPUT:

manager_id
NULL
101
102
104
103

PART C: Updating Records (Client_master Table)

1.Insert five records into the Client_master table.

```
SELECT * FROM Client_master;
```

OUTPUT:

ClientNO	Name	City	State	Bal_due
C121	Rohit Sharma	Mumbai	Maharashtra	3000
C122	Ananya Verma	Delhi	Delhi	6200
C123	Karan Mehta	Pune	Maharashtra	4800
C124	Sneha Iyer	Bangalore	Karnataka	7500
C125	Amit Khanna	Chandigarh	Punjab	5200

2.Find the names of clients whose bal_due > 5000.

```
SELECT Name
FROM Client_master
WHERE Bal_due > 5000;
```

OUTPUT:

Name
Ananya Verma
Sneha Iyer
Amit Khanna

3.Change the bal_due of ClientNO "C123" to Rs. 5100.

```
UPDATE Client_master
SET Bal_due = 5100
WHERE ClientNO = 'C123';
```

OUTPUT:

ClientNO	Name	City	State	Bal_due
C123	Karan Mehta	Pune	Maharashtra	5100
NULL	NULL	NULL	NULL	NULL